

Personalized Natural Language Content Generation using Adversarial Methods on Pre-trained Neural Recommendation Systems

Kairo Morton
mortonk@mit.edu

Kavya Anbarasu
kavyaa@mit.edu

Laena Tieng
laenat@mit.edu

1 Introduction

Deep learning approaches for natural language generation (NLG) have been applied to a variety of tasks such as question and answering, summarization, and image captioning. However, research on utilizing NLG to create personalized novel content is a much less explored topic. The ability to generate personalized content provides great value to both content creators and consumers. More concretely, creators can generate content suggestions based on information about their user base which, in turn, provides users with content better tailored to their interests. Current methods rely on a rigid subset of user characteristics such as user vocabulary, personality, and other identifiers (Krishna et al., 2018; Mairesse and Walker, 2010; Ahmad et al., 2022) to tailor the generated natural language for a specified audience. Unfortunately, these methods are not easily applicable to the problem of generating personalized textual content given arbitrarily detailed information about a user. In contrast, we propose a framework for personalized natural language content generation that uses a pre-trained recommendation system as an optimization tool to guide the generation. In summary, our key contributions are as follows:

- We develop a method for using adversarial attack techniques on neural recommendation systems to generate novel personalized content.
- We create a framework for evaluating the performance of a system built to generate personalized content.

2 Related Works

In our approach, we draw upon many well-established techniques including:

Natural Language/Content Generation Natural language generation is the process of producing free-form natural language text to meet a communicative goal (Dong et al., 2021). This communicative goal varies by task, including but not limited to summarization, question and answering, story generation, machine translation and image captioning. These goals can be grouped into two major categories: data-to-text and text-to-text generation. Nonetheless, the same set of modern models and training methods such as RNNs, transformers, and large scale pre-training have been applied successfully to solve these tasks (Dong et al., 2021).

Our task of interest, personalized natural language content generation, falls at the intersection of the two aforementioned categories as it requires generating content from arbitrary user data which may include text. Compared to many of the previously stated tasks, personalized natural language content generation still remains a widely unexplored field with previous works relying on outdated methods (Mairesse and Walker, 2010) or using limited and non-learnable user features for personalization (Polozov et al., 2015). We will attempt to expand this area of research in a way that is more generalizable by combining the following methods to generate personalized content.

Neural Recommendation Systems As summarized in Wu et al. (2021), modern neural recommendation systems take the form of a variety of model types such as collaborative filtering, content-enriched, and temporal/sequential models. These models are classified as such primarily based on what use they make of data pertaining to users, items, or context. Collaborative filtering models are the most researched area and ignore context, instead focusing on users' interaction history. Content-enriched models incorporate additional side information on top of what is used for collaborative filtering models whereas temporal or se-

quential models, also referred to as context-aware models, base predictions on context, item, and user data (Wu et al., 2021).

We will use a content-enriched (content-based) recommendation model NRMS (see §3.1) to optimize our inputs because we aim to base the generations on the article topics users have actually interacted with in order to create personalized text tailored to each given user.

Adversarial Attacks Adversarial methods to perturb neural networks inputs are a common technique that have been successfully used to encourage model prediction error through gradient-based optimization. In response to these attack methods, work has been done to train neural networks on these perturbed inputs leading to a more robust predictive models. These attack and defense methods have been repeatedly demonstrated successfully in both the speech and image domains (Pandey and Wang, 2018; Akhtar et al., 2021).

However, this type of attack is much harder to perform on text-based predictive models due to the discrete nature of the text input making these models non-differentiable with respect to their inputs. To overcome this, current methods include a Gradient Based Distributional Attack (Guo et al., 2021) where the search for adversarial inputs is done over an adversarial distribution and similarity to a specified input text is enforced through BERTScore (Zeng et al., 2019). This method provides a means of generating a variety of personalized text given that we are able to sample from a distribution and includes constraints that help enforce fluency in the loss calculation (see §3.2).

3 Methods

The main objective of a recommendation system is to indicate which items (eg. articles, products, videos, webpages, etc.) a given user is most likely to interact with. For the purposes of our work we follow a popular formulation of neural recommendation systems as a function $S(\text{user}, \text{item})$ that takes user and item features as input and produces a score representing how likely that user is to interact with that item. This function S can be broken down into three key elements: a user encoder, item encoder and a score function. The user encoder is a neural network that takes specific user data and produces an embedding vector for that user. Similarly, the item encoder is neural network that takes item related data as input and outputs an embed-

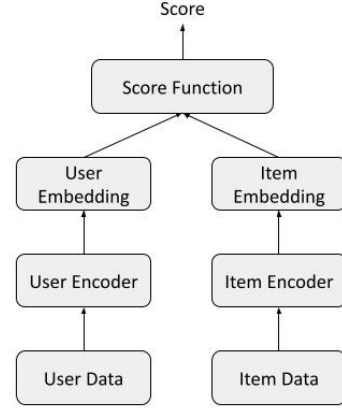


Figure 1: Recommendation system

ding vector for that item. Finally, the score function takes these two vector representations as input and returns a score predicting whether or not the user will interact with a given item.

Current recommendation systems have shown success in matching users to existing items such as products, videos, etc. but don’t typically leverage this information to generate new content. As such, given a neural recommendation system trained on user interactions with text data, we aim to utilize the techniques proposed in Guo et al. (2021) to learn a distribution of adversarial texts for any given user that maximizes the score outputted by the recommendation system (see §3.2). However, we plan to modify this method by removing the BERTScore similarity loss component while learning the adversarial distribution. In doing so, the process of generating a set of personalized novel text for a given user reduces to sampling from the aforementioned distribution. While we don’t expect the text sampled from this distribution to be factually correct, we anticipate that it accurately captures the preferences of the user. Thus, enabling content creators to leverage these generations to better understand their users’ possible interests.

3.1 NRMS

The recommendation system that we are using to optimize the content generator is called Neural News Recommendation with Multi-Head Self-Attention (NRMS) (Wu et al., 2019) which has been trained on our dataset of interest (see §3.3). It consists of a news and user encoder. The news encoder learns news representations from news titles using multi-head self-attentions to model interactions between words. The user encoder learns user

representations using multi-head self-attentions to model the relatedness between their previously clicked news titles. The score function is a dot product between the user and item embedding to indicate the likelihood of the given user interacting with the given item.

3.2 Modified GBDA

The method we will use to generate personalized natural language content is a modified Gradient-Based Distributional Attack (GBDA) (Guo et al., 2021). Typically, GBDA optimizes an adversarial distribution P_Θ such that any piece of text z sampled from this distribution minimizes a three component loss consisting of an adversarial component, fluency component, and a similarity component. This loss can be formalized as follows:

$$\mathcal{L}(\Theta) = \mathbb{E}_{z \sim P_\Theta} L_{\text{adv}}(z, y; h) + \lambda_1 L_{\text{fluency}}(z) + \lambda_2 L_{\text{sim}}(z, x)$$

where h is the text classification model being attacked, x is the original input to h , and $h(x) = y$. For text data the sampling operation $z \sim P_\Theta$ is non-differentiable because z is discrete; as such, the GBDA method extends the loss components above to take probability vector inputs directly sampled from a Gumbel-softmax distribution parameterized by Θ . Further, the fluency and similarity losses are both computed using the same pretrained language model. However, the fluency loss is computed as the negative log likelihood of the text sample z whereas the similarity score is computed using the BERTScore metric over the contextualized embeddings of z .

In this work we modify the loss function specified above by removing the similarity loss component L_{sim} and using the NRMS neural recommendation system for the model h . As such, in order to ensure that the adversarial sample text z is personalized we define the adversarial loss L_{adv} for a specific user embedding e_u as follows:

$$L_{\text{adv}} = -\log \left(\sigma \left(\frac{e_u^T e_z}{\|e_z\|_2 \|e_u\|_2} \right) \right)$$

where e_z is the news embedding of z produced by the NRMS News Encoder. Finally, for the fluency loss we follow the original GBDA paper and utilize a pretrained GPT-2 language model to compute the negative log likelihood.

3.3 Dataset

In order to pre-train a neural recommendation system and evaluate our methodology we will rely on the Microsoft News Dataset (Wu et al., 2020). In total, this dataset represents approximately 160k English news articles and 15 million impression logs generated by 1 million users. It consists of (user, item) interaction pairs where the items are news articles and the interactions are clicks obtained from the Microsoft News logs. More specifically, this dataset includes user IDs, a list of news articles each user was displayed, and whether or not the each article was clicked, in addition to relevant information about the article (title, category, subcategory, etc.).

Specifically, we train the NRMS model using a BERT tokenizer and the small version of the MIND dataset. Our version of the model achieves a AUC score of 65.23 which is comparable to the results from the original paper.

3.4 Evaluation Metrics

To measure the performance of the natural language generation model we optimize for a given user we plan to use a combination of automated and human evaluation methods to determine whether the generated text is in fact personalized and coherent. To quantify these concepts, personalized news article titles are generated for a small subset of the users present in the test set of the Microsoft News Dataset. These generated titles can then be compared for semantic similarity to the other titles a given user has interacted with using BERTScore. In doing this automated comparison we can curate a set of similar article titles that we know the user has interacted with for further manual inspection to evaluate the following important characteristics: personalization, fluency, and coherence.

4 Experiments

4.1 Baseline Results

To test the quality of our methods against a simple solution, we established a baseline using the 176 billion parameter BLOOM language model (et al., 2022). We used a small subset of the MIND test set consisting of 100 users and developed a prompt for the BLOOM model to generate a novel personalized news article title based on a randomly selected subset of previously clicked article titles. The generated text is then evaluated for similarity with the

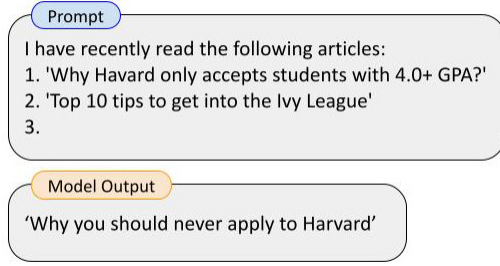


Figure 2: Example of prompt and output with BLOOM

articles that the user has interacted with before using the F1 BERTScore metric. For each candidate title generated by the language model, we compute the F1 BERTScore using all other titles that the user has interacted with before as references. The personalization score of the aforementioned candidate generation then becomes highest F1 score across all of the references. Using this evaluation schema, the BLOOM language model achieved an average BERTScore of 0.8604. These results indicate that large language models are able to achieve reasonable results and we expect that our framework will be able to create even more personalized and novel content.

4.2 Experiments

Aside from comparison with the baseline results, we are primarily investigating two aspects of the model parameters to better produce more highly rated items:

- The number of iterations of gradient descent.
- The weight of the fluency component in the loss function.

We are running this framework on 100 users for each experiment. We will be investigating two values for number of gradient descent iterations ($G=50, 100$) and three values for the lambda parameter weighting the fluency loss ($P=0.005, 0.01, 0.1$).

5 Results

The table in fig. 3 displays the performances for the different model parameters of interest. When the fluency loss is weighted less, we see that fewer gradient descent iterations improves the recommendation system's outputted scores. However, under these conditions the BERTScore decreases. On the

Results		
Model Parameters	BERTScore	Rec. System Score
$G=50, P=0.005$	0.8009	5.119
$G=100, P=0.005$	0.8030	4.708
$G=50, P=0.01$	0.8020	5.339
$G=100, P=0.01$	0.8079	4.373
$G=50, P=0.1$	0.8118	4.751
$G=100, P=0.1$	0.8139	5.772

Figure 3: Performance scores for different model parameters

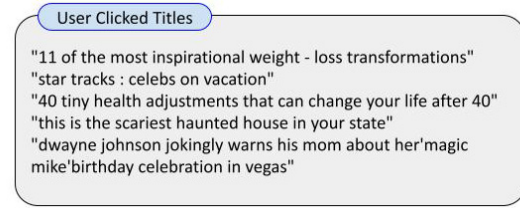


Figure 4: Clicked article history examples for User 23

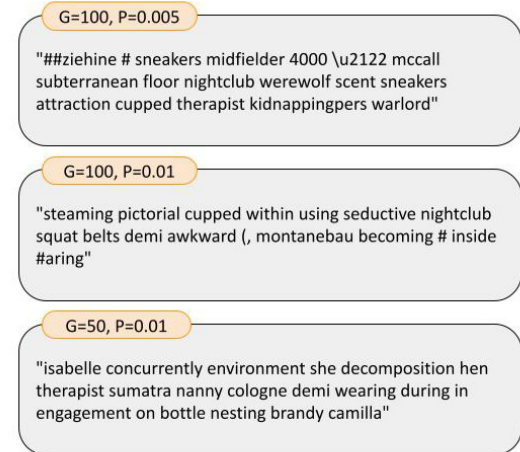


Figure 5: Generated article titles given varying model architecture's for User 23

other hand, when we drastically increased the fluency loss weight, increasing the amount of gradient descent iterations improved both metrics.

If we recall, the recommendation system score is simply just a metric to evaluate how likely the system is to recommend a given title— the higher the metric, the more likely it is to recommend the content. As such, when we raise the weight of the fluency component in the loss, the text sampled from the adversarial distribution is more likely to not include semantic information related to the user’s click history. Given this for higher fluency loss more gradient descent iterations are required to achieve high recommendation scores as more time is needed to optimize the distribution for both the adversarial and fluency objectives. As such, when we weight the fluency score more ($P = 0.1$) and have a large number of gradient descent iterations, we achieve our highest score on the recommendation system.

For BERTScore, we believe a higher number of gradient descent iterations yields a higher score because our model is able to converge upon a better optimum distribution for a given user. As such, the perturbations are perceived as better approaching in similarity to the other article titles despite possibly being more incoherent.

As shown in fig. 4 and fig. 5, the generated content for users is largely incoherent. However, when comparing the personalized generated titles to the article titles the user has previously clicked on some connections are observed. For example, the words “werewolf” and “attraction” appear in the first generated title displayed in fig. 5 which are related to the clicked title “this is the scariest haunted house in your state”.

6 Conclusion and Future Work

A tool from generating personalized content would be greatly beneficial for content creators, companies, and consumers alike who would be able to either foster or receive better-tailored experiences online. From the above results it is clear that while this methodology does not out perform the large language model baseline, it is highly dependent on the quality of recommendation system. Given that the NRMS model only attains an AUC score of 65.23 on the MIND dataset it is likely that significant improvements to the robustness and accuracy of the neural recommendation system used to personalize the generated content will result in greater

performance of our Modified GDBA approach.

Future work on models of this nature could explore extending this approach for guided natural language generation. This would involve utilizing the similarity loss component from the original GDBA approach to optimize the generated text to be similar to a user specified string. In doing so, the generated natural language would be a variation of the user specified text but personalized using the given recommendation system.

7 Impact Statement

One primary concern related to generating natural language content based on user data is that it could act as an echo chamber that only exacerbates narrow-minded and dangerous views. An extreme example would be in the case where the user data of interest comes from members of a hate group. Using extremely prejudiced and harmful topics of interest in the text data as source for the content generation, the model could then output further discriminatory content that only magnifies the issue.

Another issue that could result is the harmful misuse of the model’s generated topics as complete misinformation. While the model generates novel content, users that are not careful could take the model’s generations at face value and use that text on social media or other online platform and then, perhaps inadvertently, spread absolutely fake news. It’s important to emphasize that the model’s text generation is not a reflection of existing information that is true, but rather it is a novel generation without any form of fact checking behind it. Misunderstanding or misuse of this sort of tool provides a dangerous means of spreading harmful misconceptions, polarizes groups of people, and assists in fostering a distrust of online sources.

References

- Rangina Ahmad, Dominik Siemon, Ulrich Gnewuch, and Susanne Robra-Bissantz. 2022. [Designing personality-adaptive conversational agents for mental health care](#). *Information Systems Frontiers*.
- Naveed Akhtar, Ajmal Mian, Navid Kardan, and Mubarak Shah. 2021. [Threat of adversarial attacks on deep learning in computer vision: Survey II](#). *CoRR*, abs/2108.00401.
- Tevan Le Scao et al. 2022. [Bloom: A 176b-parameter open-access multilingual language model](#).
- Chenhe Dong, Yinghui Li, Haifan Gong, Miaoxin Chen, Junxin Li, Ying Shen, and Min Yang. 2021.

A survey of natural language generation. *CoRR*, abs/2112.11739.

Chuan Guo, Alexandre Sablayrolles, Hervé Jégou, and Douwe Kiela. 2021. [Gradient-based adversarial attacks against text transformers](#). *CoRR*, abs/2104.13733.

Kundan Krishna, Aniket Murhekar, Saumitra Sharma, and Balaji Vasani Srinivasan. 2018. [Vocabulary tailored summary generation](#). In *Proceedings of the 27th International Conference on Computational Linguistics*, pages 795–805, Santa Fe, New Mexico, USA. Association for Computational Linguistics.

François Mairesse and Marilyn A. Walker. 2010. [Towards personality-based user adaptation: psychologically informed stylistic language generation](#). *User Modeling and User-Adapted Interaction*.

Ashutosh Pandey and Deliang Wang. 2018. [On adversarial training and loss functions for speech enhancement](#). In *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 5414–5418.

Oleksandr Polozov, Eleanor O’Rourke, Adam M. Smith, Luke Zettlemoyer, Sumit Gulwani, and Zoran Popovic. 2015. Personalized mathematical word problem generation. In *Proceedings of the 24th International Conference on Artificial Intelligence, IJCAI’15*, page 381–388. AAAI Press.

Chuhan Wu, Fangzhao Wu, Suyu Ge, Tao Qi, Yongfeng Huang, and Xing Xie. 2019. [Neural news recommendation with multi-head self-attention](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 6389–6394, Hong Kong, China. Association for Computational Linguistics.

Fangzhao Wu, Ying Qiao, Jiun-Hung Chen, Chuhan Wu, Tao Qi, Jianxun Lian, Danyang Liu, Xing Xie, Jianfeng Gao, Winnie Wu, and Ming Zhou. 2020. [MIND: A large-scale dataset for news recommendation](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 3597–3606, Online. Association for Computational Linguistics.

Le Wu, Xiangnan He, Xiang Wang, Kun Zhang, and Meng Wang. 2021. [A survey on neural recommendation: From collaborative filtering to content and context enriched recommendation](#). *CoRR*, abs/2104.13030.

Wenhuan Zeng, Abulikemu Abuduweili, Lei Li, and Pengcheng Yang. 2019. [Automatic generation of personalized comment based on user profile](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics: Student Research Workshop*, pages 229–235, Florence, Italy. Association for Computational Linguistics.